

Moran process

1. At each step of a Moran process ...
 - (a) every individual reproduces once
 - (b) the population size grows by one
 - (c) the least fit individual always dies
 - (d) one individual reproduces (proportional to fitness) and one is chosen uniformly to die
2. The fixation probability of a type is ...
 - (a) the probability the population eventually consists entirely of that type
 - (b) the probability the type dies out
 - (c) the fraction of the population it starts with
 - (d) its fitness divided by the average fitness
3. Under neutral drift (a mutant of the same fitness as residents), a single mutant fixes with probability ...
 - (a) $1/N$
 - (b) 0
 - (c) $1/2$
 - (d) 1
4. A Moran process with two types has ...
 - (a) two absorbing states: all of one type or all of the other
 - (b) no absorbing states
 - (c) an absorbing state at every population split
 - (d) exactly one absorbing state
5. For an advantageous mutant with relative fitness $r > 1$, as $N \rightarrow \infty$ the fixation probability tends to ...
 - (a) $1 - 1/r$, which is bounded away from 1
 - (b) 1
 - (c) $1/N$
 - (d) 0
6. The relative fitness r of a mutant is ...
 - (a) the size of the population
 - (b) the number of mutants
 - (c) the probability it dies

- (d) its fitness divided by the resident fitness
7. In a game-based Moran process, the fitness of an individual depends on ...
 - (a) only its own type
 - (b) the composition of the whole population (its payoff in the current state)
 - (c) nothing; all fitnesses are equal
 - (d) the round number
 8. Selection in the Moran process acts through ...
 - (a) the population size
 - (b) the birth step: fitter individuals are more likely to reproduce
 - (c) neither step
 - (d) the death step: fitter individuals are more likely to die
 9. The death step of the Moran process chooses an individual ...
 - (a) that just reproduced
 - (b) uniformly at random
 - (c) with the lowest fitness
 - (d) proportional to fitness
 10. The two-type fixation formula uses the ratios γ_i defined as ...
 - (a) the fitness of type 1
 - (b) the population size
 - (c) $p_{i,i-1}/p_{i,i+1}$, the down-step over up-step probabilities
 - (d) $1/N$
 11. Under strong selection ($w = 1$) ...
 - (a) the process never fixes
 - (b) fitness equals the raw payoff, so payoff differences act at full strength
 - (c) all fitnesses are equal
 - (d) the population grows
 12. The boundary conditions for the fixation probability are ...
 - (a) $\rho_0 = \rho_N = 1/N$
 - (b) $\rho_0 = 1$ and $\rho_N = 0$
 - (c) $\rho_0 = \rho_N = 1/2$
 - (d) $\rho_0 = 0$ and $\rho_N = 1$

13. A single mutant is favoured by selection when its fixation probability satisfies ...
- (a) $\rho_1 > 1/N$
 - (b) $\rho_1 = 0$
 - (c) $\rho_1 = 1$
 - (d) $\rho_1 < 1/N$
14. The Moran process is repeated until ...
- (a) a fixed number of steps have passed
 - (b) the population doubles
 - (c) every individual has reproduced once
 - (d) only one type remains in the population
15. A disadvantageous mutant ($r < 1$) has a fixation probability that is ...
- (a) exactly 1
 - (b) above the neutral value $1/N$
 - (c) exactly $1/2$
 - (d) below the neutral value $1/N$
16. The two-type fixation probability is $\rho_i = \frac{1 + \sum_{j=1}^{i-1} \prod_{k=1}^j \gamma_k}{1 + \sum_{j=1}^{N-1} \prod_{k=1}^j \gamma_k}$, where γ_k equals ...
- (a) the mutant fitness alone
 - (b) $1/N$
 - (c) $f_R(k)/f_M(k)$, the resident-to-mutant fitness ratio
 - (d) $f_M(k)/f_R(k)$
17. In a constant-fitness process with mutant fitness r and resident fitness 1, the ratio γ_k is ...
- (a) r for every state k
 - (b) different in every state
 - (c) r^{-1} for every state k
 - (d) equal to $1/N$
18. Weak selection in the Moran process corresponds to a selection intensity w that is ...
- (a) close to 0
 - (b) close to 1
 - (c) negative
 - (d) greater than N

19. Throughout a Moran process the total population size ...
- (a) shrinks over time
 - (b) grows by one each step
 - (c) fluctuates randomly
 - (d) stays constant at N , since each birth replaces a death
20. Neutral drift is the special case of the Moran process in which the relative fitness is ...
- (a) $r > 1$
 - (b) $r = 0$
 - (c) $r < 1$
 - (d) $r = 1$